Machine learning methods have recently emerged as powerful tools to tackle quantum many–body problems, which are characterized by an exponentially growing number of degrees of freedom with increasing number of particles that makes traditional computational methods progressively expensive. A major recent development is the Neural Network Quantum States (NQS) framework, first introduced by Carleo and Troyer in 2017. Here, the variational optimization problem of finding the wavefunction for a many–body system is efficiently translated into a machine learning one, providing a flexible scheme which has already found applications across several areas of physics.

In this thesis, we investigate the use of Restricted Boltzmann Machines (RBMs) as variational ansätze for fermionic systems in a three dimensional discretized box. We validate the method on few–electron atoms (helium and lithium) and develop a first extension to nuclear physics by implementing the deuteron system described by the Argonne V6 interaction.
The implementation is based on the variational Monte Carlo (VMC) framework. The key idea is to map the physical configuration into the visible layer of the RBM in the second quantization formalism, so that at each visible unit corresponds an occupation number of a site on the lattice. By assigning binary values also to the hidden units of the network, the ansätze can be written in the closed form:

$$\psi_{\boldsymbol{\theta}}(\boldsymbol{\sigma}) = \exp\left(\sum_i a_i \sigma_i\right) \prod_j 2\cosh\left(\sum_i W_{ij}\sigma_i + b_j\right)$$

where a_i, b_j are the visible and hidden biases, respectively, W_{ij} are the complex weights connecting visible and hidden units, and $\boldsymbol{\theta} = \{a, b, W\}$ is the full parameter set. As quantum wavefunctions are generally complex-valued, all parameters of the network are taken to be complex. The sampling of configurations is carried out with the Metropolis algorithm, while the optimization of the network's parameters is done using stochastic gradient descent (SGD) method and exploits such closed form of the RBM to efficiently calculate the variational derivatives. The optimization of the wave function for the ground state is based on the variational principle and proceeds by minimizing the computed expectation value of the Hamiltonian with respect to the trial wave function. The SGD algorithm does this with a simulated annealing process in which the parameters θ are kept at a fictional finite temperature and slowly cooled down.

The implemented code was first applied to study helium and lithium atoms. We initially investigated how the learning rate decay affects training. In particular, we found out that the most effective cooling schedule is characterized by an initially hot regime, namely a high learning rate that enables a broad exploration of the variational parameter space, followed by a relatively fast temperature decrease. This quickly drives the optimization near low energy regions, allowing lower final energies to be reached. Successively, we produced an in–depth analysis of the effect of the hidden layer size on training, observing that an excessively large number of hidden units significantly complicates the optimization landscape. This makes training more unstable and often leads the RBM to converge towards local minima, likely associated with higher energy eigenstates, such as excited atomic states.

Using the best wave function found, we investigate the radial density function $\rho(r)$, the radial distribution function $P(r) = 4\pi r^2 \rho(r)$, which is the probability of finding an electron between distances r and $r + dr$ from the nucleus, and the radial pair distribution function $g(r)$ for two electrons at distance r. All plots reproduce the qualitative correct behavior for each system studied. We also provide plots of selected wavefunctions sections, obtained by fixing $N - 2$ coordinates, where $N = 3N_e$, being N_e the number of electrons, enabling a visualization of the wavefunction and symmetry effects.

Finally, we examined the main behaviors observed during the training of the RBM on the deuteron, highlighting both the regimes in which the model performs consistently and those where it encounters difficulties. At the end of optimization, the energy either remains higher or unphysically lower with respect to its expected ground state value, and we also show a representative run in which both regimes occur at different stages of training. The high energy behavior may be due to an suboptimal hyperparameter setting, in particular for the discretization of the lattice, while the lower one is probably an overfitting issue or a subtle implementation error. We then present the same spatial observables as for atomic systems, for both energy regimes. In the high–energy case, the RBM still reproduces coherent distributions for $\rho(r), P(r)$ and $g(r)$, consistent with the range of nuclear interactions. In contrast, in the low–energy regime, the sampled configurations become inconsistent with the physical system, revealing that the model is no longer capturing the nuclear structure in a correct manner.